Empirical Proof Record: VALIDATED

2026-05-24T03:30:21.007409+00:00

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Proof ID: `652217ca915b5edb3d1ea703833e5df1…`
Created: 2026-05-24T03:30:21.007409+00:00

1. Claim

Across 30 synthetic (x, y) pairs of size 100 (seed=20260518), "scipy.stats.spearmanr" and "pingouin.corr(method='spearman')" produce Spearman ρ values that agree to $\leq$ 1e-09. (RFC 0002 Phase E1: cross-framework validation at the statistical-primitive layer.)

- **Operationalisation:** Generate N (x, y) pairs where x ~ Normal(0, 1) and y = ρ_target $\times$ x + sqrt(1 - ρ_target^2) $\times$ Normal(0, 1) for a sweep of ρ_target. Compute Spearman ρ under both backends. Compute max_abs_diff = max over pairs of |ρ_scipy - ρ_pingouin|. VALIDATED iff max_abs_diff $\leq$ tolerance.
- **Threshold:** `max_absolute_spearman_difference <= 1e-09 correlation`
- **H0:** scipy and pingouin disagree on Spearman ρ by more than the floating-point tolerance — at least one library has a defect or default-behaviour drift.
- **H1:** Both libraries compute identical Spearman ρ to floating-point tolerance across every pair.

2. Verdict

VALIDATED — observed `0.0`
30 pairs checked; max |rho_scipy - rho_pingouin| = 0.000e+00 ($\leq$ 1e-09 tol)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
`spearman_scipy_vs_pingouin`	`max_absolute_spearman_difference`	`0.0`	—	scipy 1.17.1

4. Pre-registration

- Registered at: `2026-05-24T03:30:20.769240+00:00`
- Config hash: `4e41be096c5011fb2c2bed2d180c541c…`
- Data hash: `4e41be096c5011fb2c2bed2d180c541c…`

5. Signature

`309cdcabde78a7c2dd3d7af98554ddf53f2441fb5ddc62bf083341f3e97712d8`